



SMIC Process Design Kit (PDK)

Release Note & Revision History

PDK Code	SPDK18MSE_1833_OA_CDS
EDA Platform	Cadence
Title	0.18um Mixed Signal Enhanced 1.8V/3.3V Cadence OA PDK
PDK Revision	V1.11_4
Release Date	2019-1-15



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1. RELEASE NOTE

1.1	Design Platform and versions: (PDK qualified by:)	Cadence DFI version IC616-64b.500.3 Virtuoso schematic editor L, Virtuoso Layout Suite L & XL, Virtuoso ADE Environment, Spectre MMSIM12.11.188, Hspice 2010.12-SP2, Calibre 2013.4_37.29
1.2	PDK Installation	Install Utility: <pdk_unzipped_folder>/SMIC_PDK_install Please refer to “README.INSTALL” in the same folder, or “PDK Reference Manual” for more details.
1.3	PDK Reference Manual	<pdk_install_directory>/docs/PCell_Library/ SMIC_OA_CDS_Reference_Manual_18MSE_1833_V 1.11_4.pdf
1.4	Customization/ Special SPECS	N/A
1.5	Known problem & solutions	1. There may be M1.6 (min. m1 area) DRC violation in base cell layout like mosfets, diodes...etc for S/D connection or pin connection; this error would disappear when designer connects this metal out in real design.
1.6	Batch setting CDF values	Please refer to PDK Reference Manual, part-19.2, “Batch setting CDF value Utility” for details.



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1.7	Notice	1. This 018 MSE PDK (Doc NO.: TD-MM18-KA-2111), is developed based on a new platform. According to SMIC program manager, it is suggested to apply it to a brand new design. Another platform is 018 MS PDK (Doc NO.: TD-MM18-KA-2102), which is based on a phasing out platform and only will exist on SMICNOW for a certain period just for old design tape out. The SPICE model that these PDKs referred to for the 2 platforms are totally different, and are NOT compatible with each other. So it's strongly NOT suggest customer migrate design from MS to MSE platform directly. The main differences of SPICE model are: a) Model format different, one is BSIM3 and another is BSIM4 for MOS, which impact device netlist format and simulation flow & results. b) Layout structures are different on MiM and SAB resistors, which will impact simulation & silicon. For more details, please refer to SPICE model document or contact model group. If user has any special request, please contact SMIC PDK group. 2. The square number is suggested to be ≥ 5 for SAB resistor 3. W&L is suggested to be $\geq 15\mu\text{m}$ for 1fF MIM and W&L is suggested to be $\geq 8\mu\text{m}$ for 2fF MIM
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		4. We implemented parasitic diodes for MOS in this PDK, leaving the well diodes to be extracted by RC extraction flow for better precise post-simulation. The 3 parasitic diodes are: parasitic_dnwdio, (DNW / Psub) parasitic_diobpw, (Pwell / DNW) parasitic_nwdio (Nwell / Psub) <table><tr><th>MOS types</th><th>Extracted parasitic diode</th></tr><tr><td>DNW N MOS (ie.n18_dnw_ckt)</td><td>parasitic_diobpw, parasitic_dnwdio</td></tr><tr><td>PMOS DNW PMOS (ie.p12,p18, p18_dnw_ckt, p12_ckt...etc)</td><td>parasitic_nwdio,</td></tr></table> Symbol, spectre view, Hspice view, CDL views are provided for above diodes but no pcell layout views. These symbols could be used in schematic design for LVS to compare on the connectivity and parasitic diode sizes. (Need to open corresponding LVS option like below, details see LVS release note) Option12: PARASI_DIO_CMP_DNW Option13: PARASI_DIO_CMP_NW Option14: PARASI_DIO_PROPERTY_CMP_DNW Option15: PARASI_DIO_PROPERTY_CMP_NW	MOS types	Extracted parasitic diode	DNW N MOS (ie.n18_dnw_ckt)	parasitic_diobpw, parasitic_dnwdio	PMOS DNW PMOS (ie.p12,p18, p18_dnw_ckt, p12_ckt...etc)	parasitic_nwdio,
MOS types	Extracted parasitic diode							
DNW N MOS (ie.n18_dnw_ckt)	parasitic_diobpw, parasitic_dnwdio							
PMOS DNW PMOS (ie.p12,p18, p18_dnw_ckt, p12_ckt...etc)	parasitic_nwdio,							
1.8	PDK Application Note (Design Updating Notice)	This application note only supports design updating from PDK V1.11_3 to V1.11_4. Please refer to “PDK Application Note” (Chapter 4) in this document for more details.						



2. REFERENCE DOCUMENTS

Document NO.	Tech Rev	Document Title
TD-LO18-DR-2001	V1.19_REV1	0.18um Logic & Mix-signal 1.8V/3.3V Design Rule
TD-MM18-SP-2004	1.11	0.18um Mixed Signal Enhanced 1.8V/3.3V SPICE Model
TD-LO18-TF-2002	V1.19_REV1_0	0.18um Logic & Mix-signal 1.8V/3.3V process virtuoso layout technology file (OA)
TD-LO18-DC-2004	V1.19_REV1_0	0.18um Logic Mixed Signal 1.8V/3.3V Calibre DRC
TD-MM18-LC-2004	1.11_1	0.18um Mixed Signal Enhanced 1.8V/3.3V Calibre LVS
TD-MM18-XC-2018	1.11_1	0.18um Mixed Signal Enhanced 1.8V/3.3V 1P4M 1TMa1(9K) with MIM=2fF Calibre xRC
TD-MM18-XC-2025	1.11_1	0.18um Mixed Signal Enhanced 1.8V/3.3V 1P5M 1TMa1(9K) with MIM=1fF Calibre xRC



3. REVISION HISTORY

DocRev	Date	Developer	PDK Revision History																
1.11_4	2019-1-15	Tony Yang	I. Reference documents <table><tr><th>Reference Documents</th><th>Tech Rev</th></tr><tr><td>TD-LO18-DR-2001</td><td>V1.19_REV1</td></tr><tr><td>TD-MM18-SP-2004</td><td>1.11</td></tr><tr><td>TD-LO18-TF-2001</td><td>V1.19_REV1_0</td></tr><tr><td>TD-LO18-DC-2004</td><td>V1.19_REV1_0</td></tr><tr><td>TD-MM18-LC-2004</td><td>1.11_1</td></tr><tr><td>TD-MM18-XC-2018</td><td>1.11_1</td></tr><tr><td>TD-MM18-XC-2025</td><td>1.11_1</td></tr></table> II. Revision History 1. Only enhance mos CDF - sdsGateCT function.	Reference Documents	Tech Rev	TD-LO18-DR-2001	V1.19_REV1	TD-MM18-SP-2004	1.11	TD-LO18-TF-2001	V1.19_REV1_0	TD-LO18-DC-2004	V1.19_REV1_0	TD-MM18-LC-2004	1.11_1	TD-MM18-XC-2018	1.11_1	TD-MM18-XC-2025	1.11_1
Reference Documents	Tech Rev																		
TD-LO18-DR-2001	V1.19_REV1																		
TD-MM18-SP-2004	1.11																		
TD-LO18-TF-2001	V1.19_REV1_0																		
TD-LO18-DC-2004	V1.19_REV1_0																		
TD-MM18-LC-2004	1.11_1																		
TD-MM18-XC-2018	1.11_1																		
TD-MM18-XC-2025	1.11_1																		



DocRev	Date	Developer	PDK Revision History																
1.11_3	2018-10-23	Tony Yang	I. Reference documents <table><tr><th>Reference Documents</th><th>Tech Rev</th></tr><tr><td>TD-LO18-DR-2001</td><td>V1.19_REV1</td></tr><tr><td>TD-MM18-SP-2004</td><td>1.11</td></tr><tr><td>TD-LO18-TF-2001</td><td>V1.19_REV1_0</td></tr><tr><td>TD-LO18-DC-2004</td><td>V1.19_REV1_0</td></tr><tr><td>TD-MM18-LC-2004</td><td>1.11_1</td></tr><tr><td>TD-MM18-XC-2018</td><td>1.11_1</td></tr><tr><td>TD-MM18-XC-2025</td><td>1.11_1</td></tr></table> II. Revision History 1. Update PDK install script to support pure 3.3v process. 2. Enhance subcircuit mos abut function. 3. Update the TF according to the new TF file.	Reference Documents	Tech Rev	TD-LO18-DR-2001	V1.19_REV1	TD-MM18-SP-2004	1.11	TD-LO18-TF-2001	V1.19_REV1_0	TD-LO18-DC-2004	V1.19_REV1_0	TD-MM18-LC-2004	1.11_1	TD-MM18-XC-2018	1.11_1	TD-MM18-XC-2025	1.11_1
Reference Documents	Tech Rev																		
TD-LO18-DR-2001	V1.19_REV1																		
TD-MM18-SP-2004	1.11																		
TD-LO18-TF-2001	V1.19_REV1_0																		
TD-LO18-DC-2004	V1.19_REV1_0																		
TD-MM18-LC-2004	1.11_1																		
TD-MM18-XC-2018	1.11_1																		
TD-MM18-XC-2025	1.11_1																		



DocRev	Date	Developer	PDK Revision History																												
1.11_2	2017-1-8	Cf Han	I. Reference documents																												
			<table><tr><th>Reference Documents</th><th>Tech Rev</th></tr><tr><td>TD-LO18-DR-2001</td><td>1.18</td></tr><tr><td>TD-MM18-SP-2004</td><td>1.11</td></tr><tr><td>TD-LO18-TF-2001</td><td>1.18_0</td></tr><tr><td>TD-LO18-DC-2004</td><td>1.18_1</td></tr><tr><td>TD-MM18-LC-2004</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2027</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2028</td><td>1.11_1</td></tr><tr><td>TD-MM18-XQ-2029</td><td rowspan="2">1.11_0</td></tr><tr><td>~</td></tr><tr><td>TD-MM18-XQ-2034</td><td rowspan="2">1.11_0</td></tr><tr><td>TD-MM18-XQ-2036</td></tr><tr><td>TD-MM18-XQ-2039</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2044</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2048</td><td>1.11_0</td></tr></table>	Reference Documents	Tech Rev	TD-LO18-DR-2001	1.18	TD-MM18-SP-2004	1.11	TD-LO18-TF-2001	1.18_0	TD-LO18-DC-2004	1.18_1	TD-MM18-LC-2004	1.11_0	TD-MM18-XQ-2027	1.11_0	TD-MM18-XQ-2028	1.11_1	TD-MM18-XQ-2029	1.11_0	~	TD-MM18-XQ-2034	1.11_0	TD-MM18-XQ-2036	TD-MM18-XQ-2039	1.11_0	TD-MM18-XQ-2044	1.11_0	TD-MM18-XQ-2048	1.11_0
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			TD-MM18-SP-2004	1.11																											
			TD-LO18-TF-2001	1.18_0																											
			TD-LO18-DC-2004	1.18_1																											
			TD-MM18-LC-2004	1.11_0																											
			TD-MM18-XQ-2027	1.11_0																											
			TD-MM18-XQ-2028	1.11_1																											
			TD-MM18-XQ-2029	1.11_0																											
			~																												
			TD-MM18-XQ-2034	1.11_0																											
			TD-MM18-XQ-2036																												
			TD-MM18-XQ-2039	1.11_0																											
			TD-MM18-XQ-2044	1.11_0																											
			TD-MM18-XQ-2048	1.11_0																											
			II. Revision History																												
			1) Update MOS abutment procedure to support contact during some abutment condition.																												
			2) QRC file for metal option 1P6M_5Ia_1TMa4_MIM20 has been updated.																												



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1.11_1	2017-04-13	CC Du	I. Reference documents <table><tr><th>Reference Documents</th><th>Tech Rev</th></tr><tr><td>TD-LO18-DR-2001</td><td>1.18</td></tr><tr><td>TD-MM18-SP-2004</td><td>1.11</td></tr><tr><td>TD-LO18-TF-2002</td><td>1.18_0</td></tr><tr><td>TD-LO18-DC-2004</td><td>1.18_1</td></tr><tr><td>TD-MM18-LC-2004</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2027 ~ TD-MM18-XQ-2034</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2036</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2039</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2044</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2048</td><td>1.11_0</td></tr></table> II. Revision History Nothing update except for adding one notice in part-1.7 (page 4) in “SMIC_OA_CDS_Release_Note_Revision_History_18MSE_1833 V1.11 2”.pdf.	Reference Documents	Tech Rev	TD-LO18-DR-2001	1.18	TD-MM18-SP-2004	1.11	TD-LO18-TF-2002	1.18_0	TD-LO18-DC-2004	1.18_1	TD-MM18-LC-2004	1.11_0	TD-MM18-XQ-2027 ~ TD-MM18-XQ-2034	1.11_0	TD-MM18-XQ-2036	1.11_0	TD-MM18-XQ-2039	1.11_0	TD-MM18-XQ-2044	1.11_0	TD-MM18-XQ-2048	1.11_0
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TD-LO18-DR-2001	1.18																								
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1.11_0	2016-11-14	Lynn Zhang	I. Reference documents <table><tr><th>Reference Documents</th><th>Tech Rev</th></tr><tr><td>TD-LO18-DR-2001</td><td>1.18</td></tr><tr><td>TD-MM18-SP-2004</td><td>1.11</td></tr><tr><td>TD-LO18-TF-2002</td><td>1.18_0</td></tr><tr><td>TD-LO18-DC-2004</td><td>1.18_1</td></tr><tr><td>TD-MM18-LC-2004</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2027 ~ TD-MM18-XQ-2034</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2036</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2039</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2044</td><td>1.11_0</td></tr><tr><td>TD-MM18-XQ-2048</td><td>1.11_0</td></tr></table> II Revision History - Design rule and DRC keep same as PDK version 1.10_0. - SPICE model updated, impact to PCells are: - Remove 3k rhrho resistors device pcell following model card due to the process not supported. - LVS updated, for details please refer to LVS documents. - QRC updated, for details please refer to QRC documents. - Update PDK title to “0.18um Mixed Signal Enhanced 1.8V/3.3V Cadence OA PDK” - Update PDK code from “SPDK18MM_1833_Enhanced_OA_CDS” to “SPDK18MSE_1833_OA_CDS”	Reference Documents	Tech Rev	TD-LO18-DR-2001	1.18	TD-MM18-SP-2004	1.11	TD-LO18-TF-2002	1.18_0	TD-LO18-DC-2004	1.18_1	TD-MM18-LC-2004	1.11_0	TD-MM18-XQ-2027 ~ TD-MM18-XQ-2034	1.11_0	TD-MM18-XQ-2036	1.11_0	TD-MM18-XQ-2039	1.11_0	TD-MM18-XQ-2044	1.11_0	TD-MM18-XQ-2048	1.11_0
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TD-MM18-LC-2004	1.11_0																								
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TD-MM18-XQ-2039	1.11_0																								
TD-MM18-XQ-2044	1.11_0																								
TD-MM18-XQ-2048	1.11_0																								

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1.0_0	2015-5-11	Susan Wu	TD-LO18-DR-2001	9P / 1.17	1) Initial
			TD-MM18-SP-2004	0T / 1.0	
			TD-LO18-TF-2002	0P / 1.17_0	
			TD-LO18-DC-2004	2P / 1.17_0	
			TD-MM18-LC-2004	0 / 1.0_0	
			TD-MM18-XC-2016	0 / 1.0_0	
			TD-MM18-XC-2014	0 / 1.0_0	
			TD-MM18-XC-2021	0 / 1.0_0	
			TD-MM18-XC-2020	0 / 1.0_0	
			TD-MM18-XC-2015	0 / 1.0_0	
			TD-MM18-XC-2019	0 / 1.0_0	
			TD-MM18-XC-2018	0 / 1.0_0	
			TD-MM18-XC-2017	0 / 1.0_0	



4. PDK APPLICATION NOTE (Design Updating Notice)

Note: This application note only supports design updated from PDK V1.11_3 to V1.11_4

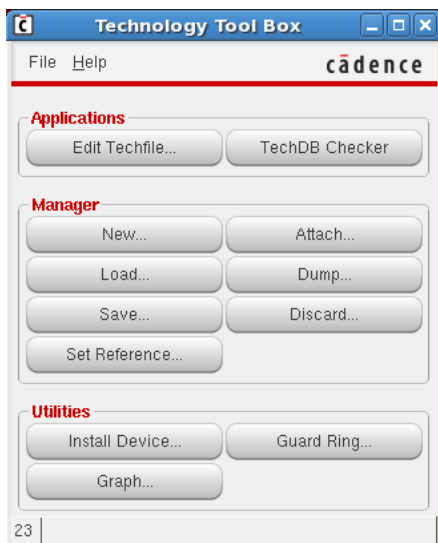
4.1 What's new in this version of PDK, impact and suggestions

Category	Update Item	Reason	Impact	Suggestion
MOS	Only enhance mos CDF - sdsGateCT function.	PCell feature enhance	No impact	NA

4.2 Update Design with new PDK

After you have installed PDK, Type **virtuoso&** in the command line, and then you can get the Command Interface Window (CIW).

Step-1: Select Tools -> Technology File Manager in CIW.



Step-2: Click Attach in Manager, choose the design library ie: "Design" and Technology library such as "smic18mmrf".

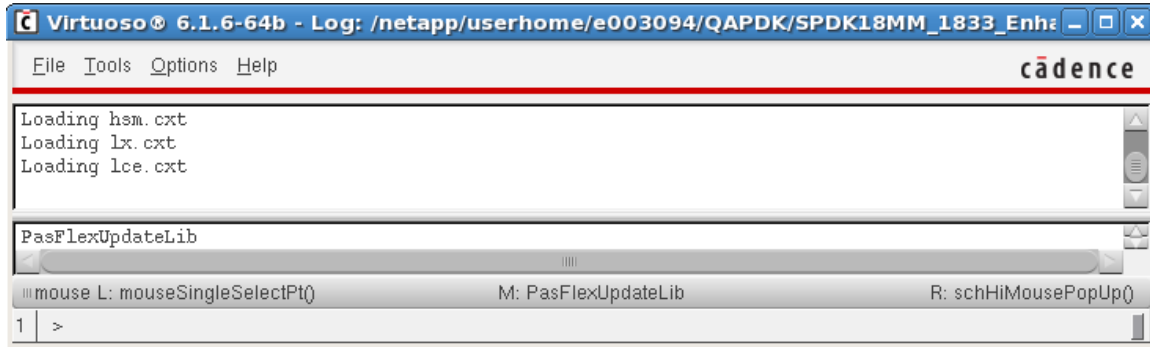




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Click ok to trigger SMIC cadence PDK.

Step-3: Then execute utility by typing “PasFlexUpdateLib” in CIW.



Step-4: Select the design library user want to update, click RUN to complete updating.

